SMART INDIA HACKATHON 2025

- **Organization-** Government of Punjab
- **PS Code-** SIH25013
- **Problem Statement Title-** Real-Time Public Transport Tracking For Small Cities
- **Theme-** Transportation & Logistics
- **PS Category-** Software
- **Team ID-**2025CSECS11
- **Team Name-**Hexabyte

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Smart Transit Tracker – Real-Time Public Transport Tracking for Small Cities

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❖Proposed Solution: Smart Transit Tracker

- Develop a real-time tracking system for buses and shared transport in small cities.
- Use GPS-enabled devices and a mobile/web application to display live vehicle location.
- Provide features like estimated arrival times, route suggestions, and seat availability.
- Enable data analytics dashboard for transport authorities to optimize routes.
- Ensure the solution is low-cost, scalable, and user-friendly.

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TECHNICAL APPROACH

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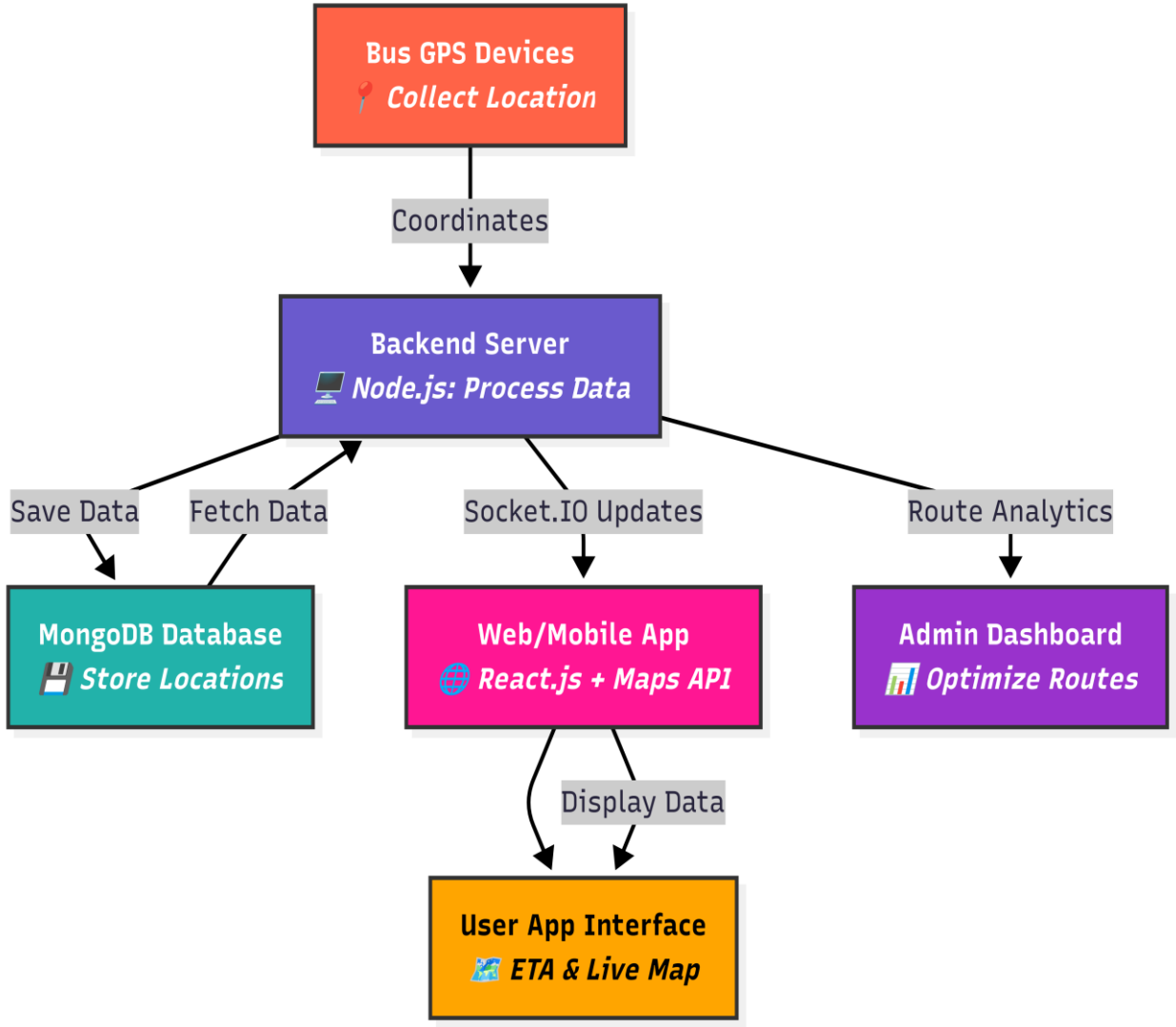
- **System Design:**
  - Use GPS devices on buses to send live location data to a backend server (Node.js + MongoDB).
  - Data is processed and displayed in real-time on a web/mobile app (React.js + Map API).

- **Implementation Methodology:**
  - Real-time updates using Socket.IO for smooth tracking.
  - Provide admin dashboard for route optimization and user app for passengers with ETA & availability.



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FEASIBILITY AND VIABILITY

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Feasibility:

- Uses existing GPS + open-source tech (Node.js, React, MongoDB)
- Scalable & cost-effective for small cities.

Challenges:

- GPS accuracy issues.
- Internet connectivity gaps.
- User adoption barriers.

Solutions:

- Assisted GPS & offline caching.
- Lightweight data transfer.
- Simple, multilingual UI.

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IMPACT AND BENEFITS

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Impact:

- Provides commuters with real-time transport updates.
- Improves efficiency of public transport in small cities.
- Helps authorities monitor and optimize routes.

Benefits:

- Saves commuter time with accurate ETAs.
- Reduces fuel usage & waiting time → eco-friendly.
- Cost-effective and scalable solution.
- Encourages digital adoption in small cities.

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RESEARCH AND REFERENCES

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- Govt. of India **Smart Mobility & Digital India initiatives**
- Research papers on **IoT & GPS in public transport**
- **Node.js, MongoDB, React.js documentation**
- **Google Maps API / Leaflet.js** references for live maps
- Case studies of **city bus tracking systems**